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Language-modulated event segmentation examined through interruption detection and boundary tapping

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Language-modulated object-state change

Intersecting Object Histories → People mentally track an object's initial and end states across its transformations when perceiving an event (Altmann & Ekves, 2019)

Event Segmentation Theory → People break down continuous activity into discrete events (Zacks & Tversky, 2001)

While English linguistic contexts regarding substantial versus zero object-state changes modulate mental representations of these changes via sentence-picture paradigms (Kang et al., 2020; Prystauka et al., 2024), relatively little is known about **whether cross-linguistic differences influence how viewers segment real-life events involving such changes in real time.**

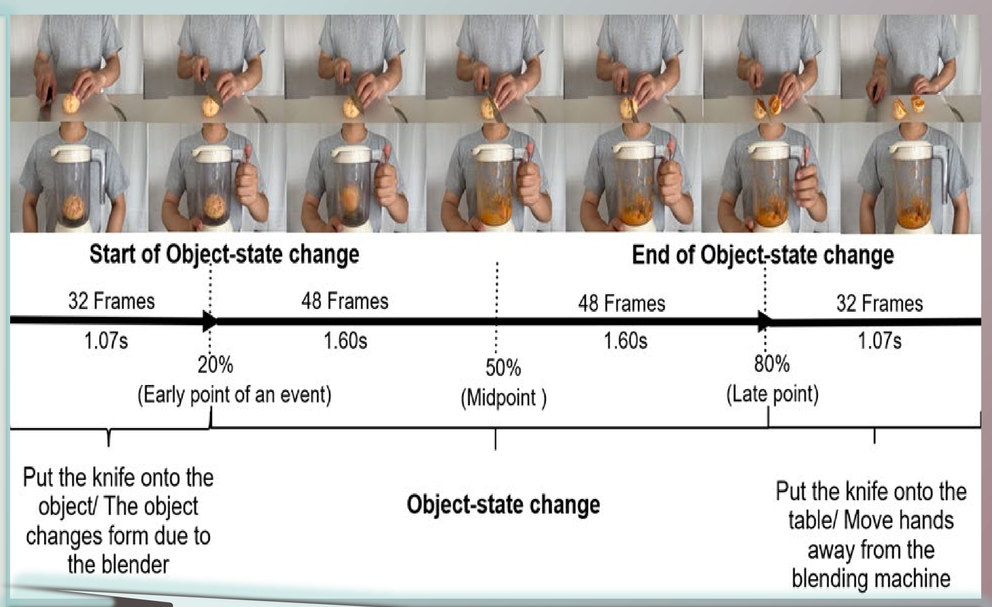
Cutting and blending events in English and Mandarin

Cutting and blending events are both object-state change events, but differ in event boundedness.

- Cutting: bounded, inherent endpoint (split into two halves)
- Blending: unbounded, no inherent endpoint (indefinite continuation)

Languages differ when encoding object-state changes.

- Verb (lexical): English resultative constructions are optional and bounded-specific; Mandarin resultative constructions are pervasive across both bounded and unbounded events. (Talmy, 2000; Chen, 2007).
- Noun (lexical): English uses obligatory determiners for individuation; Mandarin uses bare nouns as the default (Chierchia, 1998).
- Telicity (phrasal): English speakers produce more telic descriptions when describing bounded events (Ji & Papafragou, 2026).



RQ 1: Do English and Mandarin differ at the lexical and phrasal level when encoding cutting and blending events? **(Exp 1)**
RQ 2: Do lexical differences modulate internal event segmentation of cutting and blending events? **(Exp 2)**
RQ 3: Do phrasal differences modulate external event segmentation of cutting and blending events? **(Exp 3)**

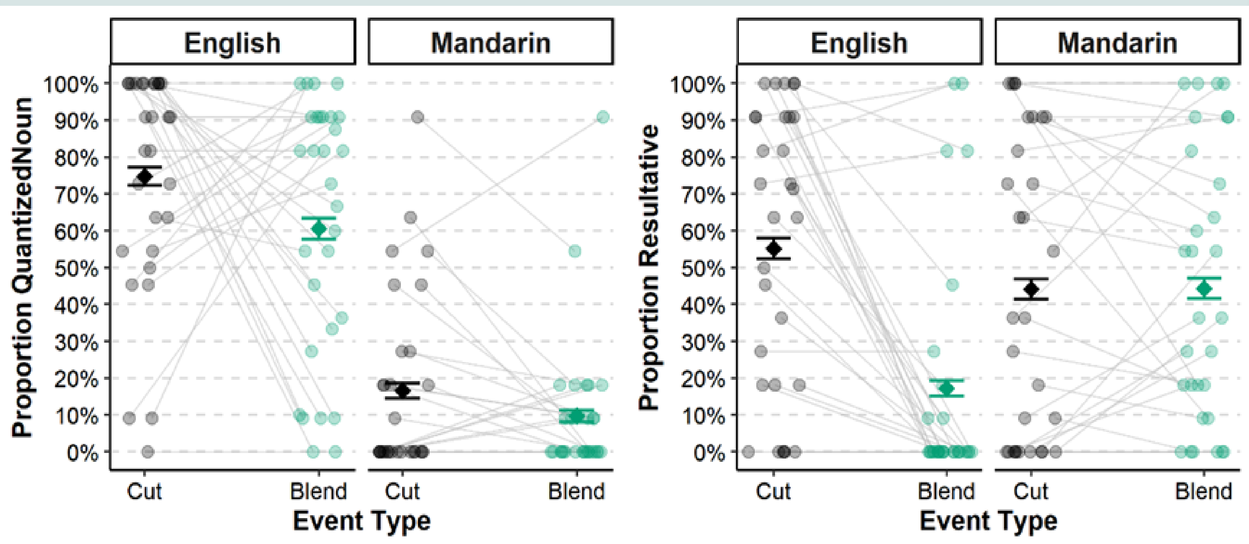
Methods:

Exp 1 Event description: Participants were asked to describe cutting and blending events as soon as they recognized the event content. Percentage data are shown in the results.

Exp 2 Interruption detection: Participants were asked to detect interruptions in the videos as soon as possible. The interruptions were inserted at the early (20%), middle (50%), and end (80%) stages of event unfolding. The accuracy data are shown in the results.

Exp 3 Boundary tapping: Participants were asked to tap the beginning and end of the object state change. End point Reaction Time (End RT) Data are shown in the results.

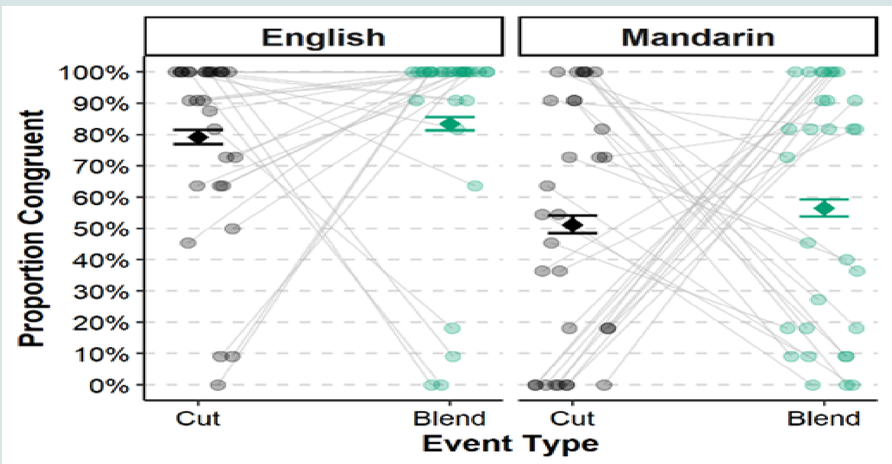
Exp 1 results at the lexical level



Lexical level:

- English speakers used **quantized nouns** significantly more frequently than Mandarin speakers.
- English speakers were significantly less likely than Mandarin speakers to use **resultative constructions** in blending events.

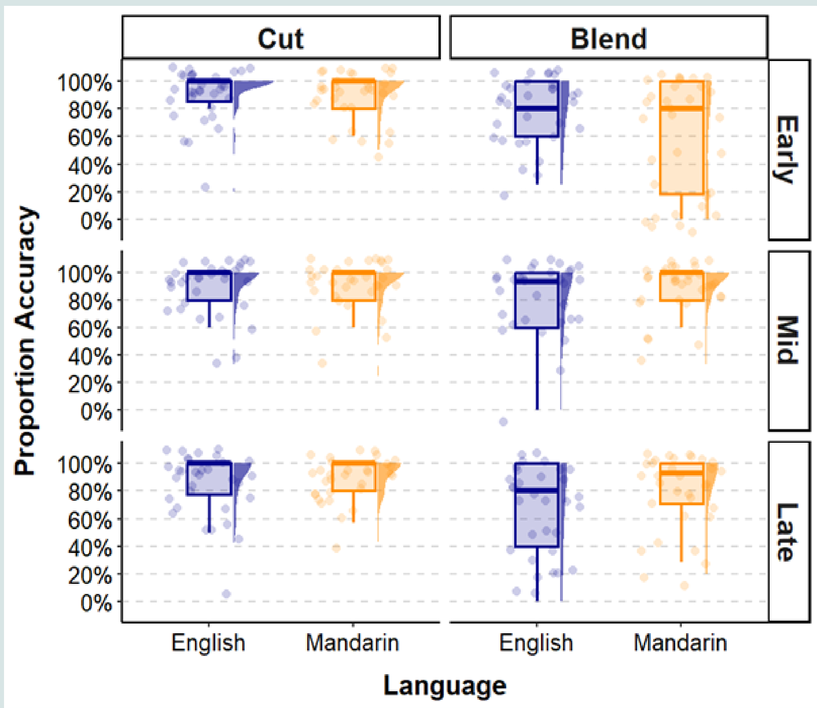
Exp 1 results at the phrasal level



Phrasal level:

- English speakers used more **telic descriptions** than Mandarin speakers when describing the bounded object-state change in cutting events.

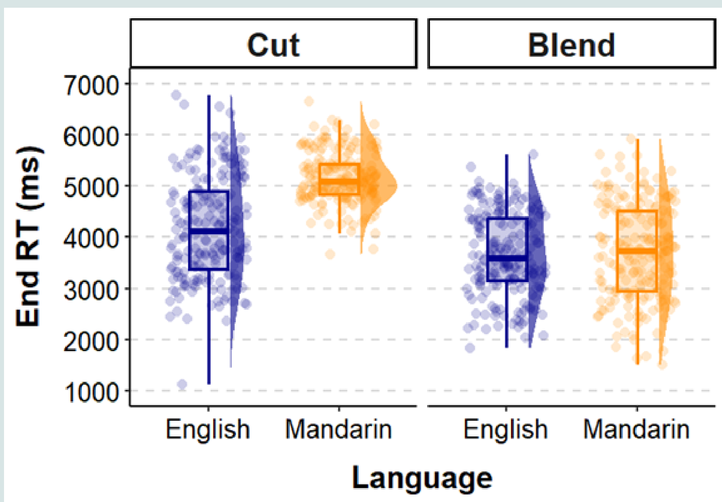
Exp 2 Internal segmentation results



Internal segmentation:

- In blending events, English speakers' higher accuracy at the **early event stage** reflects an initial-state bias driven by **quantized nouns**.
- Mandarin speakers' superior accuracy at the **late event stage** aligns with their greater use of **resultative constructions** and **the absence of obligatory individuation markers**.

Exp 3 External segmentation results



External segmentation:

- English speakers tapped **faster near the endpoint** of bounded cutting changes. They used more congruent **telic phrases** to describe cutting events than Mandarin speakers, so they were more sensitive to the end of that change in external event segmentation.

Conclusion

- Our findings support the idea that linguistic features at both the **phrasal level** (e.g., telicity) and the **lexical level** (e.g., quantized nouns and resultative constructions) influence how speakers segment object-state-change events, both **externally and internally**. This underscores how cross-linguistic differences in event encoding are associated with event segmentation.
- Furthermore, our results highlight the role of experimental paradigm: **interruption detection** vs. **boundary tapping** tasks can selectively reveal different aspects of event processing, with lexical versus phrasal cues playing distinct roles.
- Finally, real-time measures of event segmentation are particularly sensitive to subtle linguistic effects that may be overlooked by retrospective methods.

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